

Lessa?! | لسه

Smart Government Queue Tracking Platform

API Documentation

Version 1.0 — Hackathon MVP Edition | June 2026

Version 1.0 | Base URL: <https://api.lessa.app/api/v1>

1. API Conventions

Convention	Value
Base URL	https://api.lessa.app/api/v1
Protocol	HTTPS only
Format	JSON (Content-Type: application/json)
Auth	Bearer JWT in Authorization: Bearer {token} header
Pagination	Cursor-based: ?cursor={id}&limit={n}
Dates	ISO 8601 UTC: 2026-06-09T10:30:00Z
Language	Accept-Language: ar or en header; defaults to ar
Versioning	URI versioning: /api/v1/

2. Error Format

```
{
  "error": {
    "code": "QUEUE_FULL",
    "message": "No available slots for this service today.",
    "message_ar": "لا توجد خانات متاحة لهذه الخدمة اليوم.",
    "field": null,
    "details": {}
  }
}
```

Standard Error Codes

Code	HTTP Status	Description
UNAUTHORIZED	401	Missing or invalid token
FORBIDDEN	403	Insufficient role
NOT_FOUND	404	Resource not found
VALIDATION_ERROR	422	Request body validation failed
QUEUE_FULL	409	Daily capacity reached
MAX_BOOKINGS_EXCEEDED	409	Citizen has active booking
OUTSIDE_OPERATING_HOURS	409	Branch not open
CANCELLATION_CUTOFF_PASSED	409	Too late to cancel
TICKET_ALREADY_CALLED	409	Concurrent call conflict
SLOT_UNAVAILABLE	409	Race condition on last slot
SERVICE_INACTIVE	409	Service not accepting bookings
PHONE_ALREADY_REGISTERED	409	Duplicate phone on registration

ACCOUNT_LOCKED	423	Too many failed login attempts
INTERNAL_ERROR	500	Unexpected server error

3. Authentication Endpoints

POST /auth/register

Register a new citizen account

Roles: Public (unauthenticated)

```
// Request Body
{
  "phone_number": "+201234567890",
  "full_name": "Ahmed Hassan",
  "full_name_ar": "أحمد حسن",
  "language_preference": "ar"
}
```

```
// Response 201
{
  "message": "OTP sent to phone number.",
  "session_token": "otp-session-abc123"
}
```

Status Codes: 201 Created, 409 PHONE_ALREADY_REGISTERED

POST /auth/verify-otp

Verify OTP and receive tokens

Roles: Public

```
// Request Body
{
  "session_token": "otp-session-abc123",
  "otp_code": "123456"
}
```

```
// Response 200
{
  "access_token": "eyJhbGciOi...",
  "refresh_token": "rt_abc123",
  "user": {
    "id": "uuid",
    "full_name": "Ahmed Hassan",
    "role": "citizen",
    "language_preference": "ar"
  }
}
```

Status Codes: 200 OK, 401 Invalid OTP

POST /auth/refresh

Exchange refresh token for new access token

Roles: Authenticated (any role)

```
// Request Body
{ "refresh_token": "rt_abc123" }
```

```
// Response 200
{
  "access_token": "eyJhbGciOi...",
  "refresh_token": "rt_newtoken"
}
```

Status Codes: 200 OK, 401 Expired/used token

POST /auth/logout

Invalidate refresh token

Roles: Authenticated

Status Codes: 204 No Content

4. Services Endpoints

GET /services

List all active services with queue stats

Roles: Public (unauthenticated)

Query Params: ?branch_id=uuid

```
// Response 200
{
  "data": [{
    "id": "uuid",
    "code": "NID",
    "name": "National ID Renewal",
    "name_ar": "تجديد بطاقة الرقم القومي",
    "avg_handling_time_minutes": 5,
    "daily_capacity": 100,
    "current_queue_length": 23,
    "available_slots": 77,
    "load_percent": 23,
    "predicted_wait_minutes": 35,
    "is_active": true
  }]
}
```

Status Codes: 200 OK

POST /services

Create a new service

Roles: admin

```
// Request Body
{
  "branch_id": "uuid",
  "code": "PASSPORT",
  "name_en": "Passport Renewal",
  "name_ar": "تجديد جواز السفر",
  "avg_handling_time_minutes": 8,
  "daily_capacity": 80
}
```

Status Codes: 201 Created

PATCH /services/:id

Update service (including deactivate)

Roles: admin

```
// Request Body (partial update)
{
  "avg_handling_time_minutes": 6,
  "is_active": false
}
```

Status Codes: 200 OK, 409 Active bookings exist

5. Tickets (Queue Booking) Endpoints

POST /tickets

Book a new queue slot

Roles: citizen

Query Params: Header: Idempotency-Key: {uuid}

```
// Request Body
{
  "service_id": "uuid"
}
```

```
// Response 201
{
  "id": "uuid",
  "ticket_number": "NID-042",
  "daily_sequence": 42,
  "service": { "id": "uuid", "name": "National ID Renewal", "name_ar": "..."},
  "status": "booked",
  "queue_position": 18,
  "predicted_wait_minutes": 45,
  "booked_at": "2026-06-09T09:15:00Z",
  "booking_date": "2026-06-09"
}
```

```
}

```

Status Codes: 201 Created, 409 QUEUE_FULL / MAX_BOOKINGS_EXCEEDED / OUTSIDE_OPERATING_HOURS

GET /tickets/active

Get citizen's current active ticket

Roles: citizen

```
// Response 200 - Ticket object or null if no active ticket
{
  "id": "uuid",
  "ticket_number": "NID-042",
  "status": "waiting",
  "queue_position": 5,
  "predicted_wait_minutes": 12
}
```

Status Codes: 200 OK

GET /tickets/:id

Get ticket by ID with status history

Roles: citizen (own), staff, supervisor, admin

```
// Response 200
{
  "id": "uuid",
  "ticket_number": "NID-042",
  "status": "waiting",
  "queue_position": 5,
  "predicted_wait_minutes": 12,
  "citizen": { "id": "uuid", "full_name": "Ahmed Hassan" },
  "service": { "id": "uuid", "name": "National ID Renewal" },
  "booked_at": "2026-06-09T09:15:00Z",
  "checked_in_at": "2026-06-09T10:00:00Z",
  "status_history": [
    { "from": null, "to": "booked", "at": "2026-06-09T09:15:00Z" },
    { "from": "booked", "to": "checked_in", "at": "2026-06-09T10:00:00Z" }
  ]
}
```

Status Codes: 200 OK, 403 FORBIDDEN

PATCH /tickets/:id

Check in or change service type

Roles: citizen (own ticket)

```
// Check-in
{ "action": "check_in" }
```

```
// Change service
{ "action": "change_service", "new_service_id": "uuid" }
```

Status Codes: 200 OK, 409 SLOT_UNAVAILABLE

DELETE /tickets/:id

Cancel a ticket

Roles: citizen (own, before cutoff), admin (any)

```
// Response 200
{ "message": "Ticket cancelled successfully.", "ticket_number": "NID-042" }
```

Status Codes: 200 OK, 409 CANCELLATION_CUTOFF_PASSED

GET /tickets/:id/status

Lightweight polling endpoint (SSE fallback)

Roles: citizen (own), public

```
// Response 200
{
  "ticket_id": "uuid",
  "ticket_number": "NID-042",
  "status": "waiting",
  "queue_position": 3,
  "predicted_wait_minutes": 8
}
```

Status Codes: 200 OK

6. Queue Operations (Staff) Endpoints

POST /queue/call-next

Call the next ticket for staff's assigned window

Roles: staff

```
// Request Body
{ "window_id": "uuid" }
```

```
// Response 200
{
  "ticket": {
    "id": "uuid",
    "ticket_number": "NID-042",
    "citizen": { "id": "uuid", "full_name": "Ahmed Hassan" },
  }
}
```

```

    "service": { "id": "uuid", "name": "National ID Renewal" },
    "queue_position": 1
  },
  "grace_expires_at": "2026-06-09T10:35:00Z"
}

```

Status Codes: 200 OK, 204 Queue empty, 409 TICKET_ALREADY_CALLED

PATCH /queue/tickets/:id/resolve

Mark ticket as done, skipped, or no-show

Roles: staff (own window), supervisor (any), admin

```

// Request Body
{
  "resolution": "done",
  "notes": "Optional staff note"
}

```

Status Codes: 200 OK — Updated ticket object

POST /queue/tickets/:id/recall

Re-call a skipped ticket (supervisor override)

Roles: supervisor, admin

```

// Response 200 — Ticket transitions back to "called"; audit log created

```

Status Codes: 200 OK

7. Windows Endpoints

GET /windows

List all windows with current status

Roles: supervisor, admin, staff, public (status only)

Query Params: ?branch_id=uuid

```

// Response 200
{
  "data": [{
    "id": "uuid",
    "window_number": 1,
    "label": "Window 1",
    "label_ar": "نافذة",
    "status": "open",
    "assigned_staff": { "id": "uuid", "full_name": "Sara Mohamed" },
    "current_ticket": { "id": "uuid", "ticket_number": "NID-015" },
    "service": { "id": "uuid", "name": "National ID Renewal" },

```

```
    "transactions_today": 12
  ]]
}
```

Status Codes: 200 OK

PATCH /windows/:id

Update window status or staff assignment

Roles: supervisor, admin

```
// Request Body
{
  "status": "open",
  "assigned_staff_id": "uuid"
}
```

Status Codes: 200 OK

8. Notifications Endpoints

GET /notifications

Get user's notifications

Roles: citizen

Query Params: ?is_read=false&limit=20&cursor=...

```
// Response 200
{
  "data": [{
    "id": "uuid",
    "type": "turn_approaching",
    "title": "Your turn is approaching!",
    "title_ar": "دورك اقترّب!",
    "body": "You are 3 positions away. Ticket NID-042.",
    "body_ar": "أنت على بُعد 3 أرقام. تذكرة NID-042.",
    "is_read": false,
    "created_at": "2026-06-09T10:20:00Z",
    "ticket_id": "uuid"
  }],
  "unread_count": 2,
  "pagination": { "next_cursor": "...", "has_more": false }
}
```

Status Codes: 200 OK

PATCH /notifications/:id/read

Mark notification as read

Roles: citizen (own)

Status Codes: 200 OK

PATCH /notifications/read-all

Mark all notifications as read

Roles: citizen

Status Codes: 200 OK

9. Reports Endpoints

GET /reports/daily-summary

Daily summary report for a given date

Roles: supervisor, admin

Query Params: ?date=2026-06-09&service_id=uuid (optional)

```
// Response 200
{
  "date": "2026-06-09",
  "summary": {
    "total_booked": 145,
    "total_served": 112,
    "total_no_show": 18,
    "total_cancelled": 15,
    "avg_wait_minutes": 22.4,
    "avg_handling_minutes": 5.1
  },
  "by_service": [{
    "service_id": "uuid",
    "service_name": "National ID Renewal",
    "total_booked": 80,
    "total_served": 62,
    "total_no_show": 10,
    "avg_wait_minutes": 25.1
  }]
}
```

Status Codes: 200 OK

GET /reports/peak-hours

Hourly booking/serving distribution

Roles: supervisor, admin

Query Params: ?date=2026-06-09&service_id=uuid

```
// Response 200
```

```
{
  "date": "2026-06-09",
  "hourly_data": [
    { "hour": 8, "booked": 18, "served": 14 },
    { "hour": 9, "booked": 32, "served": 28 },
    { "hour": 10, "booked": 41, "served": 30 }
  ]
}
```

Status Codes: 200 OK

GET /reports/window-performance

Per-window performance for a date

Roles: supervisor, admin

Query Params: ?date=2026-06-09

```
// Response 200
{
  "date": "2026-06-09",
  "windows": [{
    "window_id": "uuid",
    "window_number": 1,
    "staff_name": "Sara Mohamed",
    "total_served": 32,
    "total_skipped": 2,
    "avg_handling_minutes": 4.8,
    "open_duration_minutes": 450
  }]
}
```

Status Codes: 200 OK

10. Predictions Endpoint

GET /predictions/wait-time

Get predicted wait for a service at current queue state

Roles: Public (unauthenticated)

Query Params: ?service_id=uuid&position=0 (0 = new booking)

```
// Response 200
{
  "service_id": "uuid",
  "tickets_ahead": 23,
  "active_windows": 2,
  "avg_handling_time_minutes": 5,
  "historical_factor": 1.1,
  "predicted_wait_minutes": 63,
  "computed_at": "2026-06-09T10:15:00Z"
}
```

Status Codes: 200 OK — predicted_wait_minutes is null on prediction failure

11. Public Display Endpoint

GET /public/display

Display board data — current/next tickets per window + announcements

Roles: Public (unauthenticated)

Query Params: ?branch_id=uuid

```
// Response 200
{
  "branch": { "id": "uuid", "name": "Cairo Central Branch", "name_ar": "فرع القاهرة المركزي" },
  "windows": [{
    "window_number": 1,
    "label": "Window 1",
    "label_ar": "نافذة ١",
    "status": "open",
    "current_ticket": "NID-015",
    "next_ticket": "NID-016",
    "service_name": "National ID Renewal",
    "service_name_ar": "تجديد بطاقة الرقم القومي"
  }],
  "announcements": [{
    "id": "uuid",
    "title": "System Maintenance",
    "title_ar": "صيانة النظام",
    "body": "Expect delays of 10-15 minutes.",
    "body_ar": "توقع تأخيرات من 10-15 دقيقة."
  }],
  "queue_summary": [{
    "service_id": "uuid",
    "service_name": "National ID Renewal",
    "waiting_count": 18,
    "predicted_wait_minutes": 45
  }],
  "as_of": "2026-06-09T10:15:30Z"
}
```

Status Codes: 200 OK

12. Admin Endpoints

GET /admin/audit-logs

Query audit logs

Roles: admin

Query Params: ?entity_type=ticket&actor_id=uuid&from=2026-06-01&to=2026-06-09&cursor=...&limit=50

```
// Response 200
{
  "data": [{
    "id": "uuid",
    "actor": { "id": "uuid", "full_name": "Sara Mohamed" },
    "action": "TICKET_STATUS_CHANGED",
    "entity_type": "ticket",
    "entity_id": "uuid",
  }]
```

```

    "old_value": { "status": "called" },
    "new_value": { "status": "done" },
    "ip_address": "41.65.1.100",
    "created_at": "2026-06-09T10:22:00Z"
  }},
  "pagination": { "next_cursor": "...", "has_more": true }
}
```

Status Codes: 200 OK

13. Real-Time: SSE Endpoint

GET /sse/queue/:serviceId

Server-Sent Events stream for real-time queue updates

Roles: Authenticated via query param: ?token=...

Query Params: Response Headers: Content-Type: text/event-stream; Cache-Control: no-cache

```

// SSE Event Types

event: queue_updated
data: {"service_id":"uuid","queue_length":20,"next_ticket":"NID-043","as_of":"..."}

event: ticket_called
data: {"ticket_number":"NID-042","window_number":1}

event: announcement
data: {"id":"uuid","title":"Delays expected","title_ar":"...", "body":"..."}

event: heartbeat
data: {"ts":"2026-06-09T10:15:30Z"}
```

Status Codes: text/event-stream; auto-reconnect handled by browser EventSource API

14. Prediction Formula Reference

Heuristic formula: predicted_wait = (tickets_ahead × avg_handling_time) / max(active_windows, 1) × historical_factor

Variable	Source	Notes
tickets_ahead	Count of tickets with status in (booked, checked_in, waiting, called) ahead in queue	Computed at request time
avg_handling_time_minutes	services.avg_handling_time_minutes	Configurable per service by admin
active_windows	Count of windows currently open and assigned to this service	Clamped to minimum of 1
historical_factor	actual_avg / configured_avg for last 30 transactions	Defaults to 1.0 if < 5 data points; clamped to 0.5–3.0

```
// Example: 8 tickets ahead, 5 min handling, 2 open windows, factor = 1.1
// predicted_wait = (8 × 5) / 2 × 1.1 = 22 minutes

// Graceful degradation: if prediction throws, API returns null for
// predicted_wait_minutes and UI shows "Estimated: N/A"
```